

Exploring the Realm of Pixels and Color Models provided a fascinating insight into the process of transforming digitized data into the colorful images we all see on a daily basis. Among other things, during my exploration, I learned that digital images are made of grids of picture elements, called pixels. Each of them has exact coordinates and a color, and all of this information together creates a whole picture. I also learned about the additive RGB color model used for displaying color images on a computer screen. This model combines three basic colors, red, green, and blue, into millions of combinations to produce a wide palette. The HSV color model, on the other hand, divides color into three components, hue, saturation, and value, which mimics human vision.

The combination of work within the framework of courses and work on two tough jobs requires the creation of an effective and well-organized plan. In order to make sure that the task was performed correctly, I chose the innovative technique of telling a story that focuses on the transformation of a color panda. The character will help me to interpret complex concepts of light mathematics and colors channels in terms of an interesting story. Having planned the project's stages properly and using advanced design software in creating a final presentation, I managed to achieve the best results of the both types without any loss of studying time.

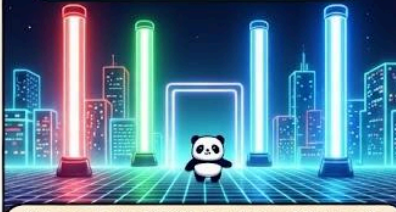
Below is my comic strip and I attached the color model visual in a separate attachment during submission.

### Meet Pao the Pixel!



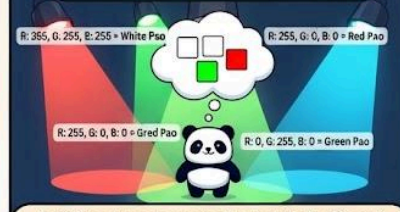
Hello! I'm Pao, the smallest unit of a digital image. Alone, I'm just a little square, but together, we make the big picture!

### Pao Enters the RGB Kingdom



Welcome to the RGB Kingdom! Everything here is made of light. Pao, you're about to see how we create color!

### Changing Colors: RGB Magic



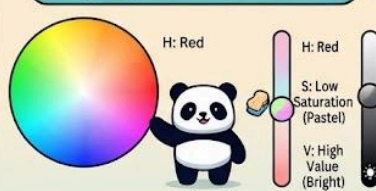
In RGB, colors are created by \*adding\* different intensities of Red, Green, and Blue light. We mix to make millions!

### Crossing to the HSV Realm



Now, let's shift to a more intuitive place: the HSV Realm! It's less about light math, more about how humans see color.

### Changing Colors: HSV Intuition



In HSV, we use three components: **HUE** (the color family), **SATURATION** (intensity), and **VALUE** (brightness). Pao can change his color effortlessly!

### Pao's Colorful World



From tiny squares to vibrant light and intuitive sliders, understanding pixels and color models unlocks the world of digital imaging. Pao loves being part of the picture!